

PostgreSQL Fundamentals Exercise

Single-Table SQL Queries and ERD Design

Student:		Date:	
Class:	SQL Fundamentals	Database:	PostgreSQL

Learning Objectives

- Write SELECT queries against a single PostgreSQL table.
- Filter rows using WHERE and comparison operators.
- Use AND, OR, NOT, BETWEEN, IN, and LIKE.
- Sort and limit query results.
- Use DISTINCT, LIMIT, ORDER BY, and basic string concatenation.
- Translate business requirements from user stories into relational ERDs.
- Identify entities, attributes, primary keys, foreign keys, relationships, and cardinality.

Naming Convention

Use lowercase, snake_case, and singular names for tables and columns. Examples: employee, support_ticket, employee_id, hire_date.

Part 1 - Single-Table SQL Queries

Scenario

You are working with a simple employee table for a small technology company. Your task is to retrieve, filter, and sort employee data. All Part 1 queries must use only the employee table. Do not use JOINS or aggregate functions.

Setup: Run the provided PostgreSQL_Fundamentals_Employee_Setup.sql script before beginning the query tasks.

Step 1 - Create the employee Table

```
CREATE TABLE employee
(
    employee_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    job_title VARCHAR(100) NOT NULL,
    department VARCHAR(100) NOT NULL,
    salary NUMERIC(10, 2) NOT NULL,
    hire_date DATE NOT NULL,
    employment_status VARCHAR(20) NOT NULL
);
```

Step 2 - Insert Example Data

```
INSERT INTO employee
```

```
(
    first_name,
    last_name,
    job_title,
    department,
    salary,
    hire_date,
    employment_status
)
VALUES
('Tony', 'Stark', 'Senior Developer', 'technology', 125000.00, '2021-03-15', 'active'),
('Peter', 'Parker', 'Junior Developer', 'technology', 72000.00, '2025-06-10', 'active'),
('Diana', 'Prince', 'Security Analyst', 'cybersecurity', 98000.00, '2022-09-01', 'active'),
('Bruce', 'Wayne', 'Security Engineer', 'cybersecurity', 115000.00, '2020-01-20', 'active'),
('Clark', 'Kent', 'Business Analyst', 'operations', 85000.00, '2023-04-11', 'active'),
('Natasha', 'Romanoff', 'Project Manager', 'operations', 105000.00, '2021-11-08', 'active'),
('Steve', 'Rogers', 'Support Specialist', 'support', 65000.00, '2024-02-14', 'active'),
('Shuri', 'Udaku', 'Software Architect', 'technology', 140000.00, '2019-07-22', 'active'),
('Wanda', 'Maximoff', 'Data Analyst', 'data', 90000.00, '2023-08-17', 'active'),
('Sam', 'Wilson', 'Support Specialist', 'support', 63000.00, '2025-01-05', 'active'),
('Carol', 'Danvers', 'Security Manager', 'cybersecurity', 130000.00, '2018-10-29', 'active'),
('Scott', 'Lang', 'Junior Developer', 'technology', 70000.00, '2024-09-12', 'inactive');
```

Student Query Tasks

1. Display all columns and all employees.
2. Display only the employee's first name, last name, and job title.
3. Display all employees who work in the technology department.
4. Display all employees whose salary is greater than 100000.
5. Display all employees whose salary is between 70000 and 100000, inclusive.
6. Display all employees who work in either the technology or cybersecurity department.
7. Display all employees who do not work in the support department.
8. Display all active employees.
9. Display all employees hired after January 1, 2023.
10. Display all employees hired before January 1, 2022.
11. Display all employees with the job title Junior Developer.
12. Display all employees whose last name begins with the letter P.
13. Display all employees whose job title contains the word Security.
14. Display all employees ordered alphabetically by last name.
15. Display all employees ordered from highest salary to lowest salary.
16. Display the five highest-paid employees.

17. Display the first name, last name, and salary of employees earning at least 90000, ordered from highest salary to lowest salary.

18. Display all unique department names.

Challenge Queries

19. Display employees who work in technology and earn more than 80000.

20. Display employees who work in either technology or cybersecurity and earn at least 100000.

21. Display active employees hired during 2024 or later.

22. Display the three lowest-paid active employees.

23. Display employee names in the format: Stark, Tony

Hint for #23: PostgreSQL uses || for string concatenation.

Part 2 - ERD Exercise: Fitness Center

User Story

As a fitness center manager, I want a database that tracks members, trainers, and training sessions so that I can see who belongs to the gym and which trainers are working with which members.

Each member has a member ID, first name, last name, email address, phone number, and membership start date.

The fitness center employs trainers. Each trainer has a trainer ID, first name, last name, email address, and specialty.

Members may schedule multiple training sessions. A trainer may conduct multiple training sessions. Each training session is conducted by one trainer and is scheduled for one member.

The system must record the date, start time, duration, and notes for each training session.

Student Task

- Identify the entities described in the user story.
- List the attributes for each entity.
- Identify a primary key (PK) for each entity.
- Identify all required foreign keys (FK).
- Draw the relationships between the entities.
- Show the cardinality of each relationship (for example, 1:M).
- Use lowercase, snake_case, singular naming.

Questions to Consider

- Can one member have many training sessions?
- Can one trainer conduct many training sessions?
- Can one training session have multiple members?
- Where should the session date and duration be stored?

ERD Workspace / Notes:

Part 3 - ERD Exercise: IT Help Desk

User Story

As an IT support manager, I want a database that tracks employees, technicians, and support tickets so that the company can manage technical problems reported by employees.

Each employee has an employee ID, first name, last name, email address, and department. Employees may submit multiple support tickets.

Each support ticket has a ticket ID, title, description, date created, priority, and status. Each support ticket is submitted by one employee.

The IT department has technicians who work on support tickets. Each technician has a technician ID, first name, last name, email address, and specialty. A technician may be assigned to multiple tickets.

For this exercise, assume that each ticket can be assigned to only one technician. A ticket may initially be created without a technician assigned.

Student Task

- Identify the entities and attributes.
- Identify all primary keys and foreign keys.
- Identify the one-to-many relationships.
- Determine which foreign key should allow NULL and explain why.
- Show relationship cardinality on the ERD.
- Use lowercase, snake_case, singular naming.

Design Question

If a support ticket has not been assigned yet, how should the relationship between the ticket and technician be represented?

ERD Workspace / Notes:

Submission Requirements

- One .sql file containing answers to all 23 Part 1 query tasks.
- Two ERD diagrams: Fitness Center and IT Help Desk.
- ERDs must show entity names, major attributes, PKs, FKs, relationships, and cardinality.
- Use lowercase, snake_case, singular naming throughout.
- Part 1 must use only the employee table; do not use JOINS.